UNIVERSITÀ DI TRENTO

Exploring data from turning experiments

Report

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1 Introduction

In this analysis of CNC turning experiments, key parameters were derived for two machining configurations: one at constant cutting speed (V_c) and the other at constant spindle speed (N). Based on the measured spindle power, the cutting power (P_c), cutting force (F_c), and specific cutting force (K_c) were calculated for each working diameter. To achieve this, the data were filtered to isolate only the segments corresponding to the effective cutting phases.

The machining parameters were:

- Work material: AISI 4135
- Diameter reduction (X-axis): from 48 mm to 24 mm (measurements from 46 mm onward)
- Machining length (Z-axis): 40 mm
- Depth of cut: $a_p = 1$ mm
- Feed rate: $f = 0.2$ mm/rev
- Constant cutting speed: $V_c = 150$ m/min
- Constant spindle speed: $N = 1200$ rpm

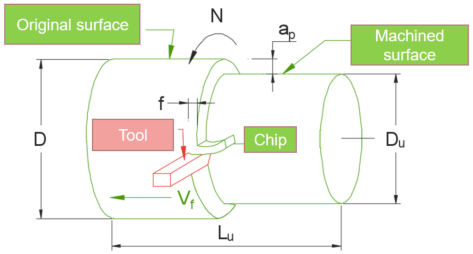


Figure 1.1. Turning setup used in the experiments.

2 Data filtering

In order to extract only the data segments corresponding to the actual cutting phases, the records were filtered by intersecting the position parameters previously mentioned (cf. *Filtering data in order to select only the cutting parts* section of the Python code). The figures below show the selected observations.

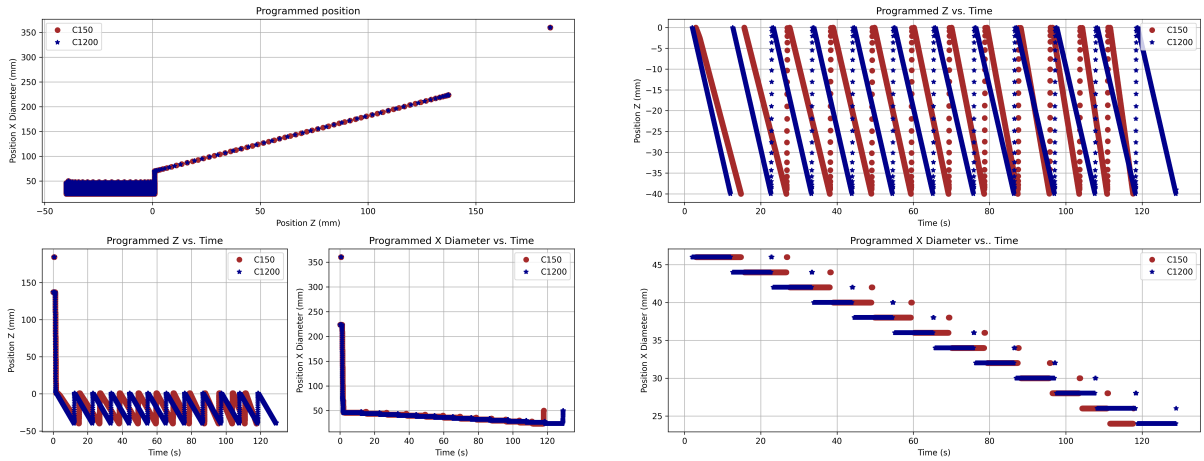


Figure 2.1. Programmed positions, respectively on the left before filtering and on the right after filtering.

Since it was necessary to consider the average power values for each diameter, the data were also filtered to remove abnormal power peaks that could affect the statistical results. To achieve this, for each diameter only the values with a deviation from the mean power lower than twice the standard deviation were retained. In other words, the signal was filtered to include only power values within the interval $[\mu_P - 2\sigma_P, \mu_P + 2\sigma_P]$ (cf. *Filtering data to avoid power peaks* section of the Python code). As a result, most of the power peaks were effectively attenuated.

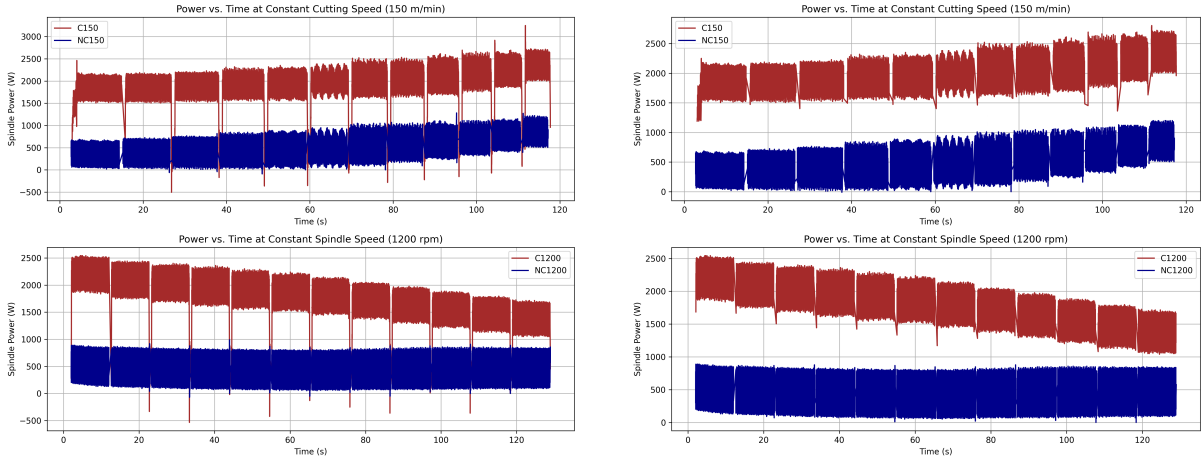


Figure 2.2. Power plots before and after the filtering.

In addition, another mask was applied to the first cutting phase at the diameter $X = 46\,\text{mm}$. As shown in the adjacent figure, in the case of constant cutting speed with chip-removal (C150), the power initially exhibits a transient increase. Since the analysis focuses on parameter estimation, only the steady-state condition was considered. Therefore, all data before $t = 4\,\text{s}$ were truncated (cf. *FC150: excluding elements until $t = 4$* section of the Python code).

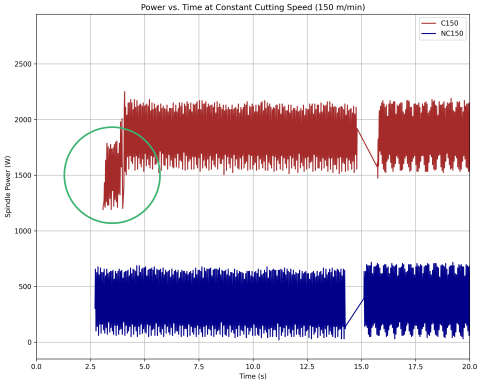


Figure 2.3. Focus on the transitory.

3 Machining Parameters Estimation

3.1 Cutting Power

Since the cutting force and the specific cutting force were derived from the cutting power, it was necessary to calculate it first. To do so, in both cases of constant cutting speed and constant spindle speed, the power component without chip removal (P_{nc}) was simply subtracted from the total power measured during machining (P_m) (cf. *Cutting Power Calculation* section of the Python code):

$$P_c = P_m - P_{nc}.$$

In Figure 3.1, the plots showing the results for each diameter are reported.

3.2 Constant Cutting Speed

After the cutting power calculation, cutting force and specific force were computed in a simple manner, considering $V_c = 150\,\text{m/min}$ constant:

$$F_c = \frac{P_c}{V_c},$$

and

$$K_c = \frac{F_c}{a_p f}.$$

3.3 Constant Spindle Speed

Similarly to the previous configuration, F_c and K_c were calculated. The only difference is that, in this case, the cutting speed V_c decreases linearly as the diameter X decreases.

$$V_c = N\pi X.$$



Figure 3.1. Cutting power for each diameter at different spindle speed conditions.

4 Analysis Results

Table 4.1. Summary statistics for cutting parameters at $V_c = 150\,\mathrm{m/min}$.

	Pc (W)				Fc (N)				Kc (MPa)			
x (mm)	$P_{c_{\min}}$	$P_{c_{\max}}$	μ_{P_c}	σ_{P_c}	$F_{c_{\min}}$	$F_{c_{\max}}$	μ_{F_c}	σ_{F_c}	$K_{c_{\min}}$	$K_{c_{\max}}$	μ_{K_c}	σ_{K_c}
	min	max	mean	std	min	max	mean	std	min	max	mean	std
46	1221.2	1907.6	1504.3	75.8	488.5	763.0	601.7	30.3	2442.5	3815.1	3008.7	151.5
44	888.2	2069.0	1485.0	340.4	355.3	827.6	594.0	136.2	1776.3	4138.1	2970.0	680.9
42	857.9	1958.0	1492.3	214.9	343.2	783.2	596.9	86.0	1715.8	3916.0	2984.6	429.8
40	878.1	2089.2	1508.4	348.3	351.2	835.7	603.4	139.3	1756.2	4178.4	3016.9	696.7
38	757.0	1877.3	1499.2	192.8	302.8	750.9	599.7	77.1	1513.9	3754.5	2998.4	385.5
36	625.8	2321.4	1515.1	452.4	250.3	928.5	606.0	180.9	1251.5	4642.7	3030.2	904.7
34	565.2	1998.4	1520.0	256.1	226.1	799.4	608.0	102.5	1130.4	3996.8	3040.0	512.3
32	1180.9	1847.0	1499.4	116.5	472.3	738.8	599.8	46.6	2361.7	3694.0	2998.9	233.0
30	968.9	1978.2	1502.1	274.8	387.6	791.3	600.8	109.9	1937.8	3956.4	3004.2	549.5
28	767.1	2230.5	1506.9	256.0	306.8	892.2	602.8	102.4	1534.1	4461.0	3013.8	512.1
26	938.6	1887.4	1495.9	63.6	375.5	754.9	598.4	25.5	1877.3	3774.7	2991.8	127.3
24	948.7	2230.5	1520.6	322.7	379.5	892.2	608.2	129.1	1897.5	4461.0	3041.2	645.3

Table 4.2. Summary statistics for cutting parameters at $N = 1200\,\mathrm{rpm}$.

	Pc (W)				Fc (N)				Kc (MPa)			
x (mm)	$P_{c_{\min}}$	$P_{c_{\max}}$	μ_{P_c}	σ_{P_c}	$F_{c_{\min}}$	$F_{c_{\max}}$	μ_{F_c}	σ_{F_c}	$K_{c_{\min}}$	$K_{c_{\max}}$	μ_{K_c}	σ_{K_c}
	min	max	mean	std	min	max	mean	std	min	max	mean	std
46	797.3	2200.2	1735.5	333.8	275.9	761.3	600.5	115.5	1379.3	3806.3	3002.4	577.5
44	1029.5	2139.7	1660.2	337.4	372.4	774.0	600.5	122.0	1861.9	3869.8	3002.6	610.1
42	979.0	2099.3	1594.0	336.5	371.0	795.5	604.0	127.5	1854.9	3977.6	3020.1	637.5
40	645.9	2079.1	1527.8	333.4	257.0	827.3	607.9	132.6	1285.1	4136.3	3039.4	663.2
38	868.0	2008.5	1461.0	330.4	363.5	841.2	611.9	138.4	1817.7	4206.0	3059.6	691.9
36	524.8	1947.9	1397.6	330.2	232.0	861.2	617.9	146.0	1160.1	4305.9	3089.4	730.0
34	696.4	1877.3	1327.8	328.3	326.0	878.8	621.6	153.7	1630.0	4393.8	3107.8	768.3
32	514.7	1766.3	1256.5	325.9	256.0	878.5	624.9	162.1	1280.0	4392.3	3124.5	810.3
30	635.9	1655.2	1184.9	327.2	337.3	878.1	628.6	173.6	1686.6	4390.6	3143.0	867.9
28	585.4	1574.5	1112.2	327.8	332.7	895.0	632.2	186.3	1663.7	4474.8	3161.1	931.5
26	292.7	1513.9	1039.8	329.1	179.2	926.7	636.5	201.5	895.8	4633.6	3182.6	1007.3
24	222.0	1443.3	967.8	331.5	147.2	957.1	641.8	219.8	736.2	4785.5	3208.8	1099.0